

Towards Ethical Research Practice: Anticipating Social Consequences of Rehabilitation Robots

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Abstract—Historically, the advent of new technology has accompanied by social issues, such as unequal access to the technology and job displacement. Rehabilitation robotics technology for post-stroke survivors is not an exception. The high costs of many of rehabilitation robots make only a few major hospitals and rehabilitation centers in urban areas viable consumers. Potentially, the robots 1) may be accessible only by those who have had access to these facilities already and 2) can replace some of the therapists in these facilities. In order to mitigate these undesirable social consequences, we draw upon Rogers' theory on the diffusion process of technological innovations, which emphasizes the role of scientists and engineers in shaping the technology throughout the whole process. We suggest that scientists and engineers address the technological needs in a way that the resulting products can assume the roles that are complementary to those of humans rather than take over them. Specifically, through the case study on InMotion ARM and Lokomat, we analyze the three intrinsic elements, *form*, *function*, and *meaning* of these robots to assess and anticipate their potential impact on society. We contend that the collective efforts of scientists and engineers may enable the deployment of rehabilitation robotics technology to be less invasive to the existing labor market as well as accessible to a larger post-stroke population. This, in turn, may build the foundation where humans and robots can coexist in various work domains.

I. INTRODUCTION

We have witnessed that technological innovations can cause negative social issues as much as they can improve the standard of living. However, scientists and engineers, who create these technologies, are rarely considered to be responsible for contemplating potential social problems and developing technologies in a way to mitigate them. The field of rehabilitation robotics is not an exception, which is the focus of this paper. Potentially, the research practice of scientists and engineers with low concern for social consequences can contribute to two significant problems if not more. The high costs of these robots could make only a few major hospitals and rehabilitation centers in urban areas viable consumers. Consequently, these robots may be accessible only by those who have had access to the medical service of these facilities already. In addition, upon the research direction chosen by scientists and engineers to develop the technology, the robots may replace therapists that are affiliated with these urban major facilities. Are scientists and engineers free from the responsibilities of these consequences? In this paper, we

attempt to address this tricky but unavoidable question: How can scientists and engineers exercise research practice to create technologies in a way that they are more responsible for the potential consequences?

To approach this question, we situate the development of robot-mediated rehabilitation technology in the context of innovation-development process illustrated in the diffusion theory proposed by Everett M. Rogers [1]. With a brief introduction to the diffusion theory of innovations, in Section II, we explain how it serves as our framework to understand the achievement in rehabilitation robotics. We emphasize that the roles of scientists and engineers are influential throughout the entire innovation-development process, especially for the earlier stages of needs acknowledgment and research and development (R&D). Section III serves as an introduction to the field of rehabilitation robotics focusing on the context of inclusion of robots into therapies for post-stroke patients. We explain the challenges that were traditionally deemed to prevent the patients from receiving necessary therapies and how the robotics community has responded to address these issues. The study objectives, questions, and designs addressed by the researchers are analyzed through a case study in Section IV. Based on this, we discuss the connection to potential job displacement and unequal access in terms of the three intrinsic elements of innovations, i.e. *form*, *function*, and *meaning*. In Section V, in conjunction with the philosophical studies which address the necessity of normative framework and ethical evaluation for robotics research, we discuss how our approach enhance ethical research practice to mitigate social issues pertaining to the use of robots, by addressing a large social context in which robot technology interacts with existing socio-economic structure.

II. DIFFUSION OF INNOVATIONS

In this study, we draw upon the diffusion theory proposed by Rogers and employ it as our framework to understand the diffusion process of innovations. He investigates the role of communication in the process that a new innovation, such as a new product, practice or idea, diffuses through a social system. In his theory, the social adoption of innovations is done through the communication among the societal members, who share information to reach a mutual understanding about the *form*, *function*, and *meaning* of innovations. Innovations are continuously shaped and often reinvented through this

exchange of information and ongoing discussion of various actors.

In his seminal work, *Diffusion of Innovations* [1], he broadens the scope of diffusion studies by considering the entire process through which an innovation is generated while traditional diffusion researches usually account that the diffusion is started after an innovation is adopted by a society. The innovation-development process consists of all the decisions, activities and their impacts that occur from recognition of a need or problem, through basic and applied research, development, and commercialization of an innovation, through diffusion and adoption of the innovation by users, to its consequences [1, p.135]. Using many examples in various countries, Rogers analyzes the dynamics that every stage has with each other and the existing socio-economic factors which shape the way the innovation is adopted in the society and everyday lives.

As a typical example, Rogers uses the invention of a hard tomato breed and the adoption of mechanized tomato harvester in Californian farms to show how problem recognition and R&D stages predetermine the consequences of the diffusion process. The introduction of the tomato-picking machine in 1964 was initiated by the researchers at the University of California Davis who anticipated potential labor shortage from the termination of Mexican bracero program through which Mexican workers came to tomato farms in California. However, they did not pay attention to the far-reaching consequences that would adversely affect human lives. The abrupt change in the location of agriculture to be more favorable to the hard tomatoes, and the unequal access to the mechanized tomato harvester significantly decreased tomato growers in California and brought large-scale unemployment. Many scholars as well as Rogers believed that those destructive social consequences might have been mitigated if the researchers designed a smaller machine that more farmers could have adopted, or at least, if they considered the social and economic impacts of farm mechanization before they invented the machine.

The example of hard tomato harvester warns scientists and engineers against their naive presumption that their decisions and research activities will always have positive effects. Although scientists and engineers do not always perform their research with particular applications in mind, some research topics, such as robot-mediated rehabilitation, are driven by specific applications and are often linked to the social consequences. Nevertheless, the study in social consequences of technological innovations has not received adequate attention by research communities. Rogers points out that the main reasons that consequences have not been adequately studied are the researchers' assumption that the consequences will be positive and the difficulty of the study itself.

Then, how do we anticipate the consequences of innovation that may not be positive for human lives? In this study, we focus on two aspects of innovations in terms of what consists of technological innovations and how they interact with the existing socio-economic structure.

According to Rogers, the intrinsic elements of innovation consist of *form*, *function*, and *meaning*. *Form* is the directly observable physical appearance and substance of an innovation. *Function* is the contribution made by an innovation to the way of life of members of a social system. *Meaning* is the subjective

and frequently unconscious perception of an innovation by members of a social system [1, p.390]. Rogers demonstrates that, through their research, scientists and engineers can influence the *form* and the *function* of final commercial products and subsequently the social consequences. Since the *meaning* emerges and changes through ongoing communication among social members, it is crucial for researchers to understand what potential users expect from new technology and what problem they may have with new technology.

However, due to the inequality within the existing socio-economic structure, the benefits of innovation are unevenly distributed among social members. Rogers emphasizes that the unequal access to new innovation may cause a gap-widening effect between those who can adopt new technology early and those who lack resources for timely adoption. Although he acknowledges that this kind of result is inevitable in some degree, his theory on innovation-diffusion process shows the importance of scientists' endeavor to consider technology that is more accessible and diffusible to a larger population. We believe that this contemplation can help individual researchers exercise ethical research practice. Collective efforts of researchers, in turn, may contribute to addressing research questions that can lead to desirable consequences for a whole society.

III. REHABILITATION ROBOTICS FOR POST-STROKE SURVIVORS

Although research communities try to use rehabilitation robots for other possible applications, such as cerebral palsy, many technologies have been developed primarily for post-stroke patients. In this section, we provide a brief overview of the clinical and social needs for post-stroke survivors and how the robotics community has responded¹.

A stroke is a rapid loss of brain cells caused by a disturbance of blood supply to the brain, which can occur due to various reasons, such as rupture of an artery or arterial blockage. The affected region of the brain is thus unable to function, which often leads to multiple chronic deficits that are severe enough to limit activities of daily living. Deficits may include difficulty in moving the limbs, understanding or producing speech and perceiving the environment [5]. Stroke is common among the elderly population, and the number of post-stroke survivors is likely to increase in future decades as the elderly population increases [6].

Post-stroke survivors need to pursue various therapies in order to alleviate their impairments. The deficits that individuals display are assessed and treated through a variety of tasks. The tasks often reinforce the activities that are performed in daily life through the repeated exercises prescribed by a therapist. The goal is to help post-stroke survivors regain functional abilities as close as possible to their abilities before the stroke.

Unfortunately, neurophysiological understanding about fundamental recovery mechanism is uncertain [2] and traditional research reports state that most of the recovery occurs within the first few months after stroke [7]–[9], which makes

¹Interested readers are referred to other review papers for more details. Langhorne et al. [2] provides an exemplary overview of stroke rehabilitation. Prange et al. [3] and Brewer et al. [4] review various rehabilitation robotic systems and their clinical effects.

insurance companies reluctant to cover the cost of extensive therapy. The financial burden on patients has decreased the length of stay for inpatient rehabilitation and has precluded patients from receiving longitudinal therapy as needed [10]. This, in turn, aggravates societal burden arising from the cost of stroke treatment and productivity loss [11], [12].

In the field of rehabilitation robotics, many research groups address this vicious cycle by developing various robotic systems that can assist/resist the movements of impaired limbs during therapy sessions and verifying their effects. These robots are expected to relieve therapists of repetitive exhausting physical labor, monitor the motor performance of patients more accurately and provide therapy sessions with reproducible exercises for longer durations at cheaper costs.

Despite the increasing variety of robots developed, the fundamental approach is the same. During therapeutic exercises, stroke patients either hold on to or place their impaired arm on the exoskeletal components of these robots, which provide resistance for high functioning patients to make the tasks more challenging and provide assistance for low functioning patients to complete otherwise unattainable movements while targets are displayed on computer screens [13]. The recent multi-site clinical test results indicate that these robots can be as effective as human therapists within the restricted types of physical therapies in both upper-extremity [14] and lower-extremity rehabilitation [15].

In short, it is being proved that, during therapy sessions, these robots can assume some of the activities that are traditionally performed by therapists. Commercialization of rehabilitation robots like InMotion Arm and Lokomat is driven by people who find the possibility of cost reduction in that therapists can see more patients in a limited time using these robots and the subsequent costs may decrease. Indeed, Wagner et al. suggest that the overall cost reduction in stroke care might be the benefit of adopting these robots [16].

IV. CASE STUDY: INMOTION ARM AND LOKOMAT

Among many robots that are being developed for the use of physical rehabilitation, we focus on InMotion ARM (Interactive Motion Technologies, Watertown, MA, USA) and Lokomat (Hocoma, Volketswil, Switzerland) since they are arguably the two most well-known products that have been introduced to the market. In this discussion, we examine how researchers defined the *function* and the *form* of their robots in their innovation-development process, and study the *meaning* to the potential adopters.

The development of InMotion ARM and Lokomat is largely motivated by the increasing number of post-stroke patients, the shortage of well-trained therapists and the reimbursement issues, such as insurance and Medicare, which place limits on both the amount of time per day that a therapist can work with a patient as well as the number of visits per calendar year the patient can receive. When responding to these societal problems, the both research groups proposed new robotic solutions that would assume the tasks that had been expected from therapists. For instance, Hogan et al. focused on the task that had been performed by the occupational and physical therapists when they reported the development of MIT-MANUS robot workstation in 1992 [17]. They noticed

that, in a typical session, therapists would manually hold the hand/arm of the patient and physically guide its movements while the patient executed various therapeutic tasks. MANUS is designed to assume this manual assistance.

A similar approach is found in the development process of Lokomat. Colombo et al. studied the task that had been carried out by physiotherapists [18]. Therapists would hold the legs of the patient and physically assist his or her leg movements. Lokomat was designed to assume this laborious manual assistance that would be otherwise done by human therapists. In this way, scientists' focus on the tasks of therapists guided their investigation on what robots can/should do in therapy, namely the *function* of robot rehabilitation technology. The *function*, thus, is defined as the ability that enables therapists to be relieved from some of activities required during therapy sessions.

In the subsequent R&D process, the researchers attempted to improve the performance of the robots so that they can better assume the identified tasks of human therapists. As seen in Buerger et al. [19] and Dipietro et al. [20], the researchers attempted to make the MANUS provide more adaptive and customized assistance to the patients. Similarly, Jezernik et al. [21] and Riener et al. [22] attempted to improve Lokomat by enabling it to automatically generate more adaptive gait patterns for the individual patients. As Riener et al. pointed out, the ideal role of rehabilitation robots would be to assist the patients' movements only as much as needed just as qualified human therapists would do. As the *function* of these robots is defined to assume what the therapists originally performed, the development of these robots necessarily lead to the unavoidable question; could these robots match or even outperform humans in the identified tasks?

In later studies, this question was verified through multi-site studies for both InMotion ARM and Lokomat. It turns out that the patients' motor improvement from the robot-mediated therapy sessions could be comparable to that from the human-delivered intensive therapy sessions [14], [15], which may lead to the overall healthcare cost reduction [16]. The researchers anticipated that the therapists would be able to prescribe hours of robot-delivered therapy to patients and to be free from exhaustive tasks that require continuous attention by adopting these robots. Eventually, the researchers expected to contribute to solving the stroke rehabilitation by enabling a single therapist to see multiple patients simultaneously using these robots.

Both InMotion ARM and Lokomat are specially-designed robot platforms that are dedicated to particular therapeutic tasks, which limits the use of these robots. These robots are quite large and need to be fixed so that they can tolerate the weight of the patients and the dynamics of their movements. The patients place their impaired limbs in the exoskeleton components of the robots and are strapped during therapy sessions. As most specially-designed devices are, the demands of InMotion ARM and Lokomat are limited. Hence, the *form* of these robots makes them too expensive to be affordable by majority of hospitals or rehabilitation centers. InMotion ARM robot costs about \$80,000 for a single unit as of 2010², and

²In Wagner et al., however, the cost for a single unit is reportedly \$230,750 [16].

a clinical institution often boasts of being the only one that possesses this robot in the state it is located [23]. As of 2010, the total expenses to obtain a single Lokomat Pro robot is \$304,000, setting aside the maintenance and repair expenses [24]. This makes only a few major hospitals and rehabilitation centers in urban areas viable consumers. Indeed, it has been reported that, in the United States, only 25 centers possess Lokomat lower-extremity treadmill robots [25]. These include Rehabilitation Institute of Chicago (Chicago, IL) and Shepherd Center (Atlanta, GA) to name a few, which are located in urban areas. Consequently, these robots may be accessible only by those who have had access to these facilities already. Those who live in suburban areas may have limited access to these robots. The *form* of these robots can naturally lead to unequal access.

There are examples that show that researchers' consideration of the accessibility and the usability of their innovations is realized through the *function* and the *form* that they define in their research process. Baxter (Rethink Robotics, Boston, MA), a factory robot, demonstrates that it is possible to produce effective yet affordable robots. The *form* of Baxter is upper-torso humanoid that comes with an intuitive programming user interface, which allows its use in a wide variety of small businesses. The *function* of Baxter can be programmed by the end users to meet the specific needs of individual business, which creates a large volume of demands. This enables a mass production and subsequently drops the price down to \$25,000 as of 2014 [26]³.

Taking another example in the computer science community, a small number of research groups explore the possible use of general-purpose service robots with the emphasis on residential use. Instead of building exoskeleton robots that are designed to perform specific therapeutic tasks, these groups attempt to make use of residential service robots that are designed to perform general household chores. For instance, Jung et al. explore the rehabilitation in residential settings through the robots in autonomous or teleoperated modes [27]–[30]⁴. In order to modulate the challenge level, the robot starts with the exercise targets that can be achieved by patients with sufficient efforts and make the targets successively more difficult, inducing increasingly larger range of motion. Matarić et al. investigate socially assistive role of the robots and attempt to encourage patients to stay in self-driven exercises by speech and gesture [32].

The *meaning* of these robots emerges through the communication process of societal members and is dependent on the context in which they are used. As the robots relieve the therapists from manual labor, hospitals and rehabilitation centers may attempt to keep the cost-effective ratio of the human therapists and the robots. For instance, Wagner et al. suggest that the overall stroke care costs could be reduced by adopting rehabilitation robots, which requires only 15 minutes

³Although Baxter may be accessible by large population, it still bears the risk of job displacement due to its low cost.

⁴It is interesting to note that Dijkers et al. proposed a similar therapeutic task using a specially-designed robotic arm along with a separate computer workstation in 1989 [31]. Now general-purpose service robots have evolved to take over the *function* of specially-designed robots. This implies that it may be possible that general service robots will become technologically capable of delivering the type of therapies that are provided by specially-designed exoskeleton robots.

of therapist time for an hour of robot-delivered therapy session [16]. It is possible that some of the therapists may be replaced with more robots for the sake of cost reduction or revenue enhancement, especially in the facilities that the adoption of these robots is underway or imminent.

V. CONCLUSION: TOWARDS ETHICAL RESEARCH

In the case study on InMotion ARM and Lokomat, we showed that the existing *form* and *function* of rehabilitation robots may potentially cause or exacerbate two social problems of the unequal access and the unemployment of therapists. We acknowledge that it is difficult to anticipate nor prevent all the negative social consequences of developing rehabilitation robotics technologies. Nonetheless, we emphasize that the activities of researchers can shape their innovations and their social consequences.

It is predicted that machines are being rapidly replace humans in many domains as robots and computer systems become capable of more and more complex tasks [33]. It is also forecast that technological transformations will dramatically change the structure of business and industry [34]. It might be inevitable that jobs in various sectors would be taken by robots and sophisticated computer systems. As overall job market becomes favorable only to those who already have resources to prepare for new economic structure, the polarization of wealth will become more serious.

However, we strongly believe that we can shape a different future where humans and robots coexist in various work domains, not replace with nor exclude from each other. When scientists design the *form* and *function* in the initial process of research, their activities and decisions are contingent upon their assumption on the human-robot relationship they expect. As theorists on philosophy of technology have explained the dynamics on society and technology development, once technologies are introduced into society, it may alter people's perception, behavior, and way of living [35]–[37]. Also, the context in which the technological innovation will be placed, the way people will use, and the meaning the technology will have in the society are intertwined with existing societal norms and values. Specifically, in the case of healthcare robotics, scientists' assumption on this reciprocity should affect how much roles and responsibilities ought to be delegated to robots and humans respectively [38].

As Winsberghe contended [39], who suggested to include ethical criteria throughout the design process of healthcare robots, envisioning an ethical research on rehabilitation robotics thus leads us to questions on fundamental value: nature of good rehabilitation and the activities we want rehabilitation robots to fulfill. For a more ethical research, we propose to build a social consensus to have more accessible rehabilitation robots where patients can benefit from both human therapists and rehabilitation robots. The next step we move forward is to develop a framework for scientists to research and evaluate the technology they are creating.

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